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EUROSYSTEM

Seasonal adjustment of economic data using TramoSeats

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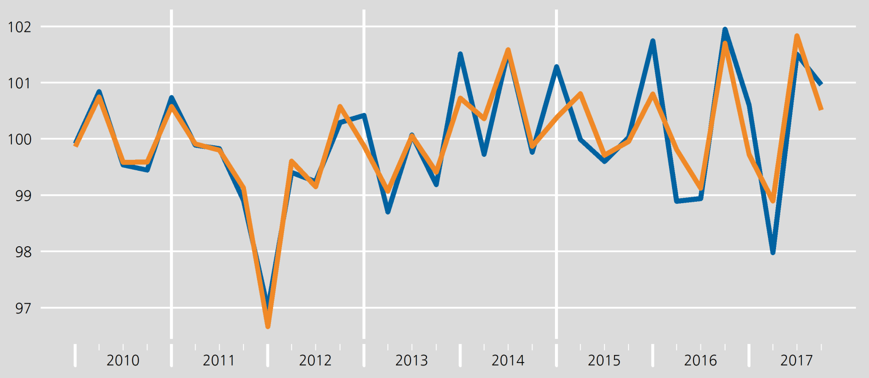
Motivation (I/III)

Spurious adjustment of X-11, quarterly series

Quarterly series

Simulated in R: rnorm(32, 100, 1)



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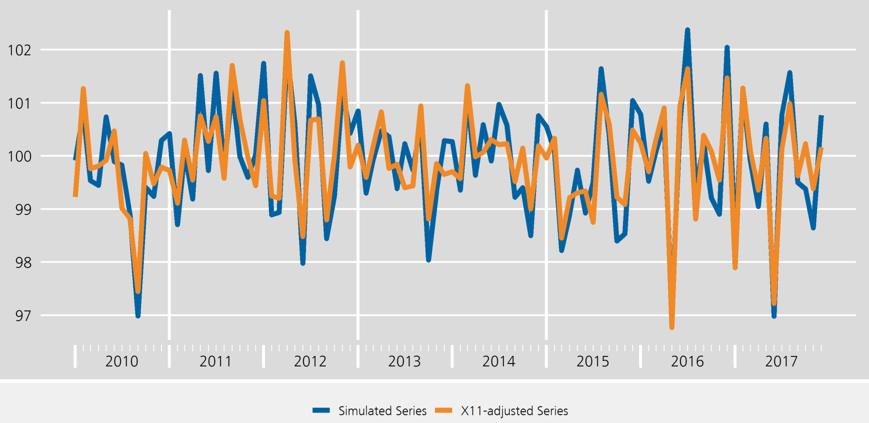
Motivation (II/III)
Spurious adjustment of X-11, monthly series

Quarterly series

Simulated in R: rnorm(32, 100, 1)



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Motivation (III/III)

Advantages of SEATS

- In theory, spurious adjustment is less of a problem
- Drawing inference is more straight-forward
- Model-based prediction of components

Seasonal adjustment procedures (I/III)

Seasonal adjustment procedure

- X-13 in JDemetra+

$$\text{X-13:} \quad \text{RegARIMA} + \text{X-11}$$

Seasonal adjustment procedures (II/III)

Seasonal adjustment procedure

- X-13 in JDemetra+

$$\text{X-13:}\quad \underbrace{\text{RegARIMA}}_{C_t} + \underbrace{\text{X-11}}_{S_t}$$

Seasonal adjustment procedures (III/III)

Seasonal adjustment procedure

- X-13 in JDemetra+

$$\text{X-13:}\quad \underbrace{\text{RegARIMA}}_{C_t} + \underbrace{\text{X-11}}_{S_t}$$

- TramoSeats in JDemetra+

$$\text{TramoSeats:}\quad \underbrace{\text{Tramo}}_{C_t} + \underbrace{\text{Seats}}_{S_t}$$

TramoSeats

Tramo = **T**ime Series **R**egression with **A**RIMA Noise, **M**issing Values and **O**utliers

Seats = **S**ignal **E**xtraction in **A**RIMA **T**ime **S**eries

Tramo vs. RegARIMA

Tramo $\approx$ RegARIMA

Tramo vs. RegARIMA

Tramo ≈ RegARIMA

Outlier adjustment

- Lower automatic CV used in Tramo
- More iterations to avoid spurious outliers in Tramo
- RegARIMA slightly better at finding correct outliers

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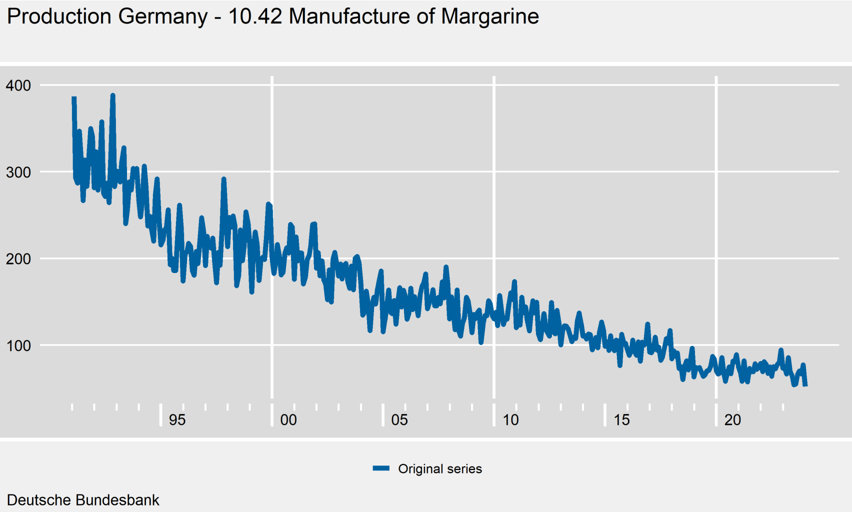
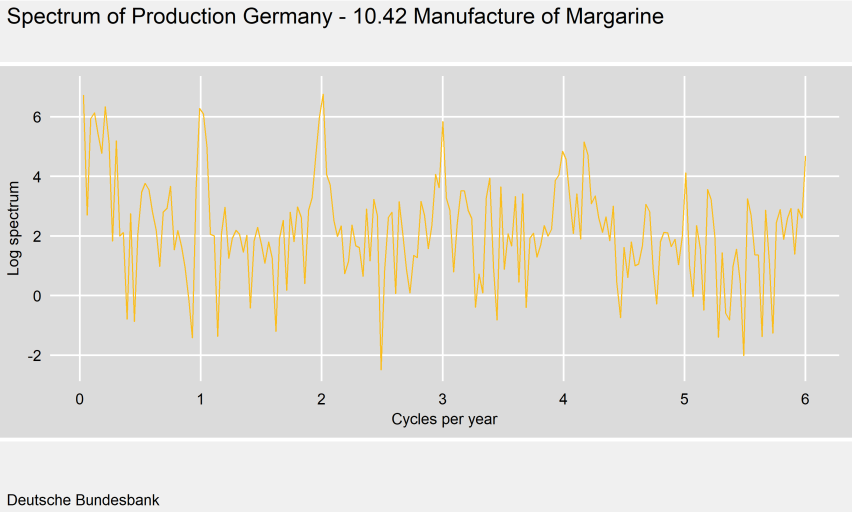
ARIMA modelling

- RegARIMA: Tends to use more exact maximum likelihood
- Tramo: Chooses airline model more frequently
- Very similar in terms of forecast accuracy

Basic concepts (I/III)



Basic concepts (II/III)



Basic concepts (III/III)

Intuition of Seasonality:

Seasonality is that part of the series that gives rise to spikes at the seasonal frequencies in the spectrum

Intuition of Seats:

SEATS separates a time series into its trend, seasonal, and irregular components by identifying their characteristic frequencies in the spectrum

Intuition of the Spectrum:

The spectrum shows how the strength of different cycles or patterns in a time series is distributed across frequencies

Assumptions

Pre-adjustment

- Outliers and calendar effects have been removed
- An ARIMA model is given

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Model

- Series = Trend + Seasonal + Transitory + Irregular
- Components are orthogonal
- Variance of Irregular component is maximized
- Ideally: $P \geqslant Q$ with P being the AR order and Q being the MA order

Seats procedure

1. ARIMA model estimation

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2. Derivation of the ARIMA models for each component

Seats procedure

1. ARIMA model estimation

2. Derivation of the ARIMA models for each component

3. Estimation of the components using Wiener-Kolmogorov filter

Seats procedure

1. ARIMA model estimation
2. Derivation of the ARIMA models for each component
3. Estimation of the components using Wiener-Kolmogorov filter
4. Diagnostic checks

1. ARIMA model estimation

Usually: Use model found by TRAMO for linearised series

2. Deriving ARIMA model for each component

$Y_t = T_t + S_t + \text{Transitory}_t + I_t$

Goal: $\varphi_i(B)x_i = \Theta_i(B)a_i$ where i refers components

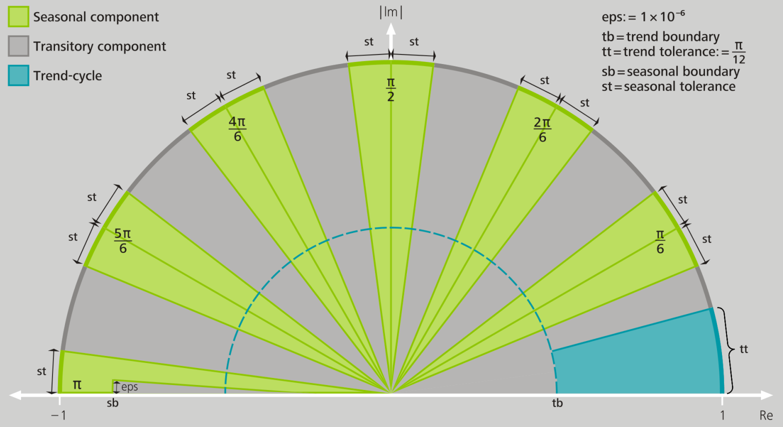
- Obtain characteristic polynomial
- Calculate roots from this polynomial
- Assign roots to components (see next graphic)
- Then you have the AR polynomials for the components
- The component MA polynomials are obtained via partial fraction decomposition

2. Deriving ARIMA model for each component

Assigning AR roots to components in monthly series

Assignment of AR factors to unobserved components in JDemetra+

Monthly time series, in radians



Deutsche Bundesbank S3IN0540.Chart

2. Deriving ARIMA model for each component
Obtaining MA polynomails

The MA terms are assigned to the models using Stochastic Partial Fraction Expansion

- We assign the MA polynomials to the components in such a way that they satisfy the following two conditions:
- Reconstruction: the series spectrum equals the sum of component spectra
- Admissibility: each component has a valid, nonnegative spectral density
- Orthogonality: components are pairwise uncorrelated
- Canonical choice: maximize the variance of the Irregular component subject to these constraints

3. Estimation of the components using WK filter

Each component is estimated using the Wiener-Kolmogorov filter

- The Wiener-Kolmogorov filter estimates each unobserved component as a weighted average of the observed time series values

- It uses both past and future observations, assigning optimal weights to minimize the estimation error for each component

4. Diagnostic checks

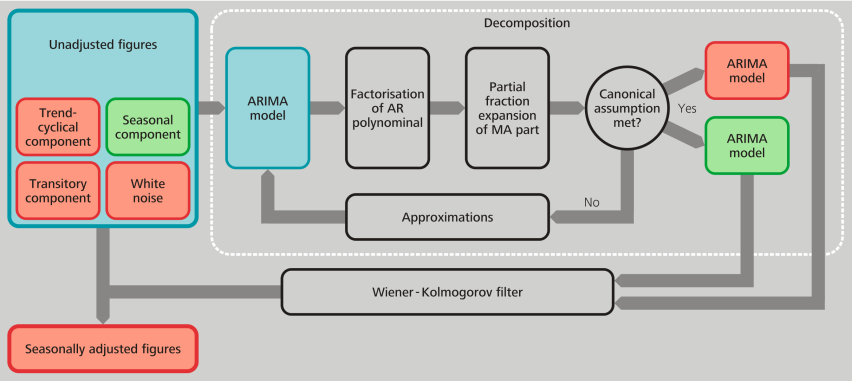
Diagnostics include

- Is ARIMA model decomposition admissible?
- Difference between theoretical estimators and the obtained estimates are compared
- Seasonality tests

Seats
Complete workflow

The basic principle of the SEATS seasonal adjustment algorithm

Workflow diagram



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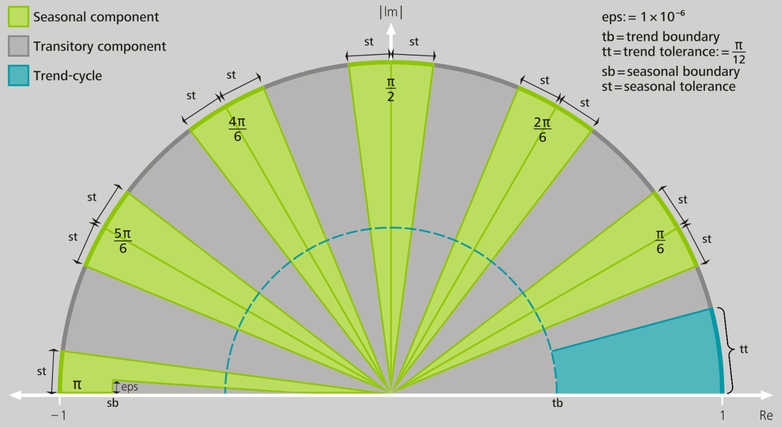
Assigning AR roots to components (I/III)

Monthly time series

Assignment of AR factors to unobserved components in JDemetra+

Monthly time series, in radians



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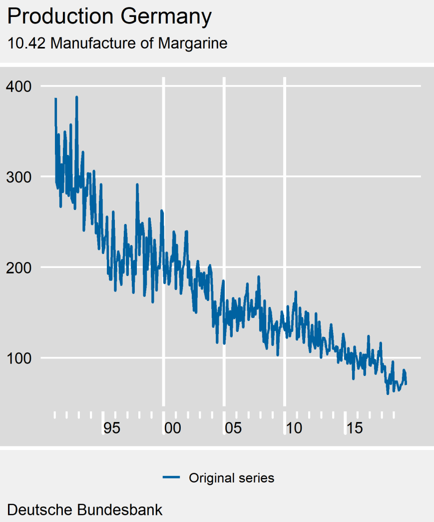
S3IN0540.Chart

Assigning AR roots to components (II/III)

An example: assigning AR to trend



Model: (3 1 1)(0 1 1) with

AR: $$(1 + 0.1587B + 0.1008B^2 - 0.2709B^3)$$

Assigning AR roots to components (III/III)

An example

Characteristic polynomial:

$$(1 + 0.1587X + 0.1008X^2 - 0.2709X^3) = 0$$

Assigning AR roots to components (III/III)

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Unit root

$$X = \begin{cases} -0.72 - 1.23i \\ -0.72 + 1.23i \\ 1.82 - 0i \end{cases}$$

Assigning AR roots to components (III/III)

An example

Characteristic polynomial:

$$(1 + 0.1587X + 0.1008X^2 - 0.2709X^3) = 0$$

Unit root

$$X = \begin{cases} -0.72 - 1.23i \\ -0.72 + 1.23i \\ 1.82 - 0i \end{cases}$$

Inverse root

$$\frac{1}{X} = \begin{cases} -0.35 + 0.60i & \leadsto Noise \\ -0.35 - 0.60i & \leadsto Noise \\ 0.55 & \leadsto Trend \end{cases}$$

Deriving MA components

Idea: Partial fraction decomposition allows us to decompose the overall ARMA filter into component filters, which yield the MA polynomials for each unobserved component.

$$x_t = \frac{\theta_t(B)}{\varphi_t(B)} a_t$$

$$x_t = \frac{\theta_T(B)}{\varphi_T(B)} \epsilon_{T,t} + \frac{\theta_S(B)}{\varphi_S(B)} \epsilon_{S,t} + \frac{\theta_I(B)}{\varphi_I(B)} \epsilon_{I,t}$$

Diagnostics (I/II)

Distribution of component, theoretical estimator and empirical estimate (stationary transformation)

Variance

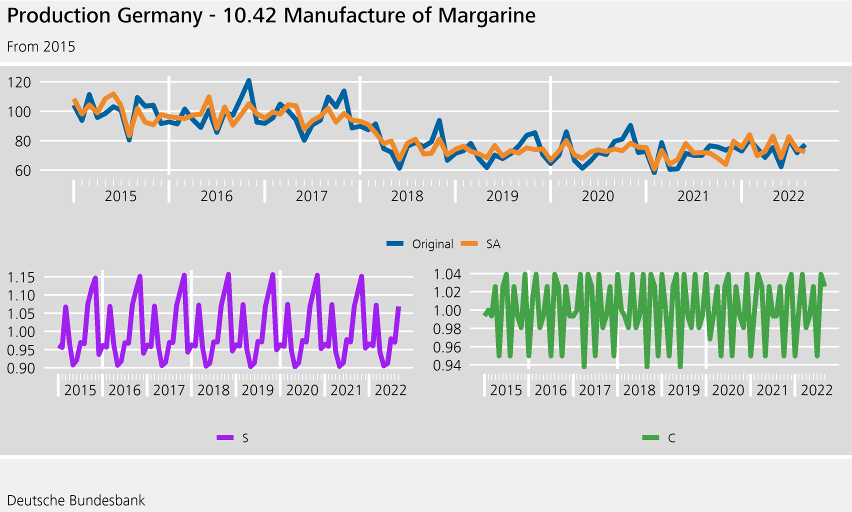
	Component	Estimator	Estimate	P-Value
Trend	0,0477	0,0037	0,0035	0,6866
Seasonally adjusted	3,8514	3,5532	3,3352	0,5586
Seasonal	0,0179	0,0014	0,0012	0,7285
Irregular	0,5717	0,4584	0,4342	0,5408

Diagnostics (II/II)

Interpretation of Variance

- V = Theoretical variance of component (Estimator)
- $\hat{V}$ = Empirical variance of component (Estimate)
- $V > \hat{V}$: Underestimation of component
- $V < \hat{V}$: Overestimation of component

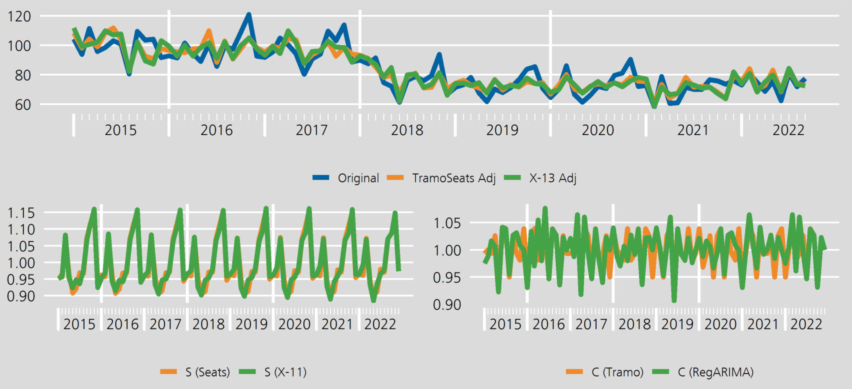
Adjustment results



Adjustment results

Production Germany - 10.42 Manufacture of Margarine

From 2015



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Seats in JDemetra+ (I/II)

SEATS	
Prediction length	-1
Approximation mode	Legacy
MA unit root boundary	0,95
Trend boundary	0,5
Seasonal tolerance	2
Seasonal boundary	0,8
Seas. boundary (uniq...	0,8
Method	Burman

Seats in JDemetra+ (II/II)

Parameter	**Description**	**Options** (*default*)
Prediction length	Forecast length	no. of periods
Approximation mode	Modification type for inadmissible models	None, *Legacy*, Noisy
MA unit root boundary	Modulus threshold for resetting MA 'near-unit-roots'	[0.9, 1] (*0.95*)
Trend boundary	Modulus threshold for assigning positive real AR roots	[0, 1] (*0.5*)
Seasonal tolerance	Degree threshold for assigning complex AR roots	[0, 10] (*2*)
Seasonal boundary	Modulus threshold for assigning negative real AR roots	[0, 1] (*0.8*)
Seasonal boundary (unique)	Same modulus threshold unique seasonal AR roots	[0, 1] (*0.8*)
Method	Estimation algorithm for the unobserved components	*Burman*, KalmanSmoother, McElroyMatrix

Take-aways for Seats

Seats …

… is ARIMA model based (AMB)

… does not require much user-interference

… includes more model based diagnostics